THE UNIVERSITY of EDINBURGH
School of Biological Sciences

COVER SHEET

Comparative and Evolutionary Genomics: Phylogeny of Mammalian Hemoglobins

In-Course Assessment 1

Student Exam Number:	B269948

Declaration	
I **have/have not** used Generative AI-based methods (e.g. ChatGPT) in the preparation of this Dissertation.	**I have not**

Word Count*	1863

*The word count should include figure and table legends but not include the tables themselves and not include references.

Methods

1. Gathering of the sequences

   Each hemoglobin protein was searched on NCBI Protein [1] and Nucleotide [2] website, each search was filtered by taxon (“Homo sapiens”) and by source databases which was set as RefSeq [3]. Then a BLASTP [4] analysis was run on each protein sequence. For mammalian genomes, the organism section of the search set was set to Mammalia and exclude homo sapiens.

   From each BLASTP [4] run, one hit was selected randomly.

   For the outgroup, random three hemoglobin subunits (HBB, HBG1, HBD) were selected. The organism filter was set to Reptilia. One sequence was selected from each run.

2. Multiple Sequence Alignment

   For protein sequences, MAFFT (v7.453) [5] alignment software was used. The maximum number of iterative refinements was set to 10000. Genafpair was used as pairing algorithm.

   For DNA sequences, EMBOSS package (EMBOSS:6.6.0.0) tranalign [6] software was used. The output file of MAFFT [5] was used as an input file.

   The visualisation of the alignment was made with Aliview software [version 1.27) [7].

3. Phylogeny reconstruction

   To find the best model and best possible solving of the phylogeny tree, IQTree software (IQ-TREE multicore version 2.2.0 COVID-edition for Linux 64-bit built Jun 1, 2022) [8] was used. Ultrafast [9] support value was set to 1000. The output files of MAFFT [5] and tranalign [6] were used as input files.

   FigTree (v1.4.4) [10] software was used to visualise and root the outputs of IQTree [8] runs on the sequences.

Results

This section of the report includes the results of the described methods.

1. Gathering of Sequences

The sequences were obtained from NCBI Protein [1] and Nucleotide [2]. The data were put on Table 1 and 2 with their accession numbers. The BLASTP [4] scores were shared on Table 3.

Table 1. The protein sequences gathered from NCBI Protein [1] and NCBI BLASTP [4].

Protein Sequence ID	Protein Sequence Name	Organism	Group in the Phylogeny Tree
NP_005323.1	hemoglobin subunit zeta (HBZ)	Homo sapiens	Ingroup
XP_024089065.1	hemoglobin subunit zeta (HBZ)	Pongo abelii	Ingroup
NP_000549.1	hemoglobin subunit alpha (HBA)	Homo sapiens	Ingroup
NP_001036091.1	hemoglobin subunit alpha (HBA)	Pan troglodytes	Ingroup
NP_000508.1	hemoglobin subunit alpha (HBA)	Homo sapiens	Ingroup
XP_030858703.1	hemoglobin subunit alpha (HBA)	Gorilla gorilla gorilla	Ingroup
NP_005321.1	hemoglobin subunit epsilon (HBE)	Homo sapiens	Ingroup
XP_025214164.1	hemoglobin subunit epsilon (HBE)	Theropithecus gelada	Ingroup
NP_000175.1	hemoglobin subunit gamma-2 (HBG2)	Homo sapiens	Ingroup
XP_034787914.1	hemoglobin subunit gamma-1 (HBG1)	Pan paniscus	Ingroup
NP_000550.2	hemoglobin subunit gamma-1 (HBG1)	Homo sapiens	Ingroup
NP_001162320.1	hemoglobin subunit gamma-2 (HBG2)	Papio anubis	Ingroup
NP_000510.1	hemoglobin subunit delta (HBD)	Homo sapiens	Ingroup
XP_010361647.1	hemoglobin subunit delta (HBD)	Rhinopithecus roxellana	Ingroup
NP_000509.1	hemoglobin subunit beta (HBB)	Homo sapiens	Ingroup
XP_004090697.3	hemoglobin subunit beta (HBB)	Nomascus leucogenys	Ingroup

XP_039364655.1	hemoglobin subunit epsilon-like (HBE-like)	Mauremys reevesii	Outgroup
XP_044880745.1	hemoglobin subunit beta (HBB)	Mauremys mutica	Outgroup
XP_067414746.1	hemoglobin subunit beta (HBB)	Emydura macquarii macquarii	Outgroup

Table 2. The CDS sequences gathered from NCBI Nucleotide [2] and NCBI BLASTP [4].

DNA CDS Sequence ID	Coordinates	DNA CDS Sequence Name	Organism	Group in the Phylogeny Tree
NM_005332.3	267-695	Homo sapiens hemoglobin subunit zeta (HBZ), mRNA	Homo sapiens	Ingroup
XM_024233297.2	306-734	PREDICTED: Pongo abelii hemoglobin subunit zeta (HBZ), mRNA	Pongo abelii	Ingroup
NM_000558.5	38-466	Homo sapiens hemoglobin subunit alpha 1 (HBA1), mRNA	Homo sapiens	Ingroup
NM_001042626.1	38-466	Pan troglodytes hemoglobin subunit alpha 1 (HBA1), mRNA	Pan troglodytes	Ingroup
NM_000517.6	38-466	Homo sapiens hemoglobin subunit alpha 2 (HBA2), mRNA	Homo sapiens	Ingroup
XM_031002843.3	92-520	PREDICTED: Gorilla gorilla gorilla hemoglobin subunit	Gorilla gorilla gorilla	Ingroup

		alpha (LOC115931067), mRNA		
NM_005330.4	56-499	Homo sapiens hemoglobin subunit epsilon 1 (HBE1), mRNA	Homo sapiens	Ingroup
XM_025358379.1	310-753	PREDICTED: Theropithecus gelada hemoglobin subunit epsilon 1 (HBE1), mRNA	Theropithecus gelada	Ingroup
NM_000184.3	54-497	Homo sapiens hemoglobin subunit gamma 2 (HBG2), mRNA	Homo sapiens	Ingroup
XM_034932023.3	16-459	PREDICTED: Pan paniscus hemoglobin subunit gamma-1 (LOC100982657), mRNA	Pan paniscus	Ingroup
NM_000559.3	54-497	Homo sapiens hemoglobin subunit gamma 1 (HBG1), mRNA	Homo sapiens	Ingroup
NM_001168849.1	1-147	Papio anubis hemoglobin subunit gamma 2 (HBG2), mRNA	Papio anubis	Ingroup
NM_000519.4	51-494	Homo sapiens hemoglobin subunit delta (HBD), mRNA	Homo sapiens	Ingroup
XM_010363345.2	196-639	PREDICTED: Rhinopithecus roxellana	Rhinopithecus roxellana	Ingroup

		hemoglobin subunit delta (LOC104662340), mRNA		
NM_000518.5	51-494	Homo sapiens hemoglobin subunit beta (HBB), mRNA	Homo sapiens	Ingroup
XM_004090649.3	176-619	PREDICTED: Nomascus leucogenys hemoglobin subunit beta (HBB), mRNA	Nomascus leucogenys	Ingroup
XM_039508721.1	43-486	PREDICTED: Mauremys reevesii hemoglobin subunit epsilon-like (LOC120387704), mRNA	Mauremys reevesii	Outgroup
XM_045024810.1	42-485	PREDICTED: Mauremys mutica hemoglobin subunit beta (LOC123374637), mRNA	Mauremys mutica	Outgroup
XM_067558645.1	33-476	PREDICTED: Emydura macquarii macquarii hemoglobin subunit beta (LOC137160199), mRNA	Emydura macquarii macquarii	Outgroup

Table 3. The BLASP [4] scores of the selected sequences.

Input Sequence	Selected Sequence	Max and Total score	E-value	Percentage Identity	Group
Homo sapiens HBZ	Pongo abelii HBZ	289	7,00E-100	99.30%	Ingroup
Homo sapiens HBA1	Pan troglodytes HBA1	286	7,00E-99	100.00%	Ingroup
Homo sapiens HBA2	Gorilla gorilla gorilla HBA	285	4,00E-98	99.30%	Ingroup
Homo sapiens HBE	Theropithecus gelada	297	5,00E-103	97.96%	Ingroup
Homo sapiens HBG2	Pan paniscus	301	2,00E-104	98.64%	Ingroup
Homo sapiens HBG1	Papio anubis	295	4,00E-102	96.60%	Ingroup
Homo sapiens HBD	Rhinopithecus roxellana	293	4,00E-101	96.60%	Ingroup
Homo sapiens HBB	Nomascus leucogenys	295	7,00E-102	97.96%	Ingroup
Homo sapiens HBB	Mauremys reevesii	229	3,00E-77	69.39%	Outgroup
Homo sapiens HBD	Mauremys mutica	224	5,00E-75	68.71%	Outgroup
Homo sapiens HBG1	Emydura macquarii macquarii	236	1,00E-79	71.43%	Outgroup

2. Analysis of aligned sequences

The sequences were aligned using MAFFT [5] and tranalign [6] software. The aligned sequences were analysed on Aliview [7].

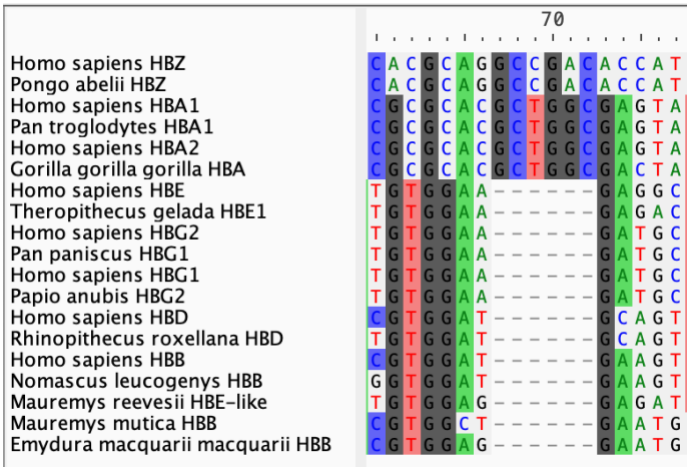


Figure 1: Figure 1 obtained while analysing the aligned DNA sequences using Aliview [7]. Sequences were taken from NCBI Nucleotide [2] and NCBI BLASTP [4], they were aligned using MAFFT [5] and tranalign [6] software.

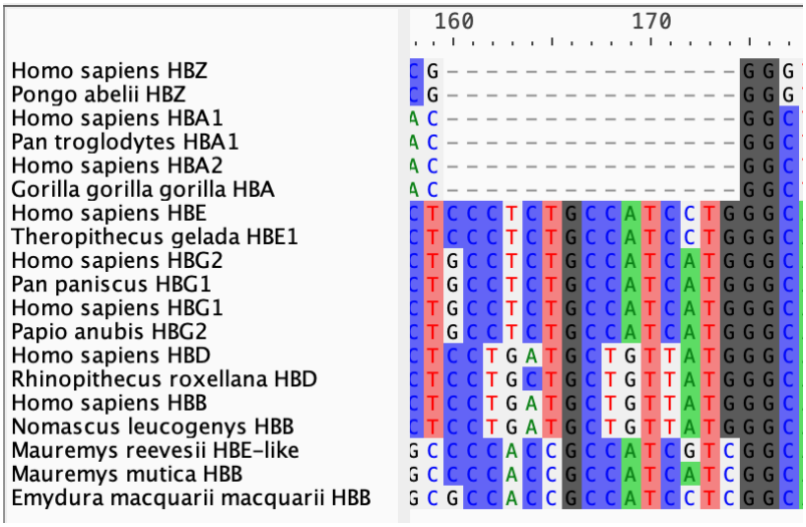


Figure 2: Figure 2 obtained while analysing the aligned DNA sequences using Aliview [7]. Sequences were taken from NCBI Nucleotide [2] and NCBI BLASTP [4], they were aligned using MAFFT [5] and tranalign [6] software.

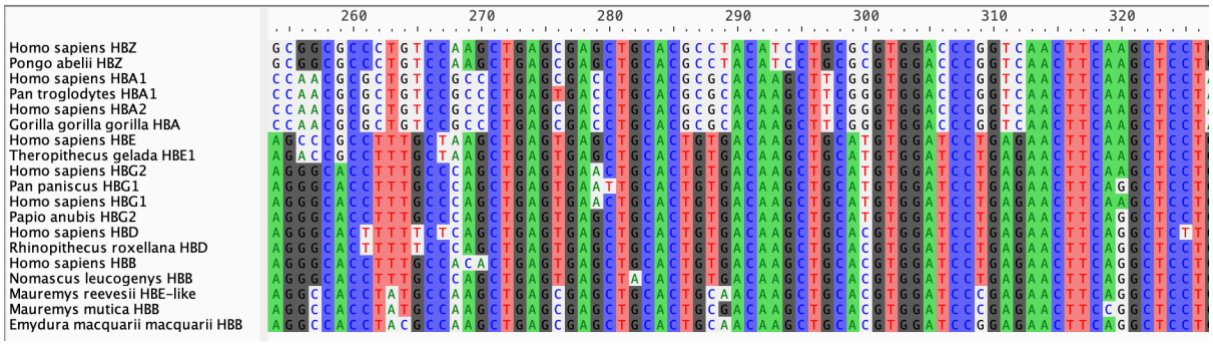


Figure 3: Figure 3 obtained while analysing the aligned DNA sequences using Aliview [7]. Sequences were taken from NCBI Nucleotide [2] and NCBI BLASTP [4], they were aligned using MAFFT [5] and tranalign [6] software.

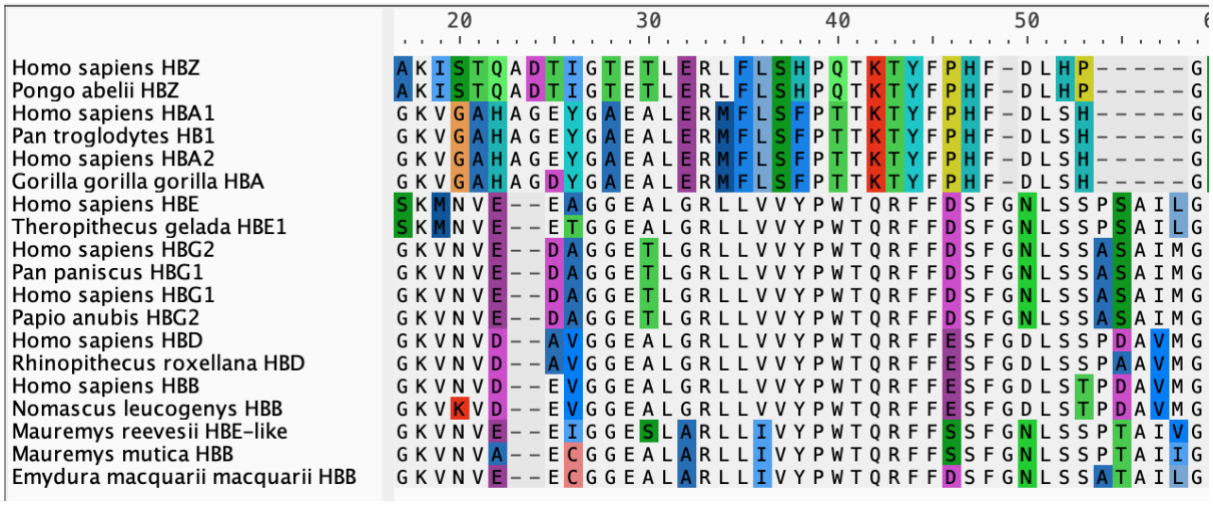


Figure 4: Figure 4 obtained while analysing the aligned protein sequences using Aliview [7]. Sequences were taken from NCBI Protein [1] and NCBI BLASTP [4], they were aligned using MAFFT [5] software.

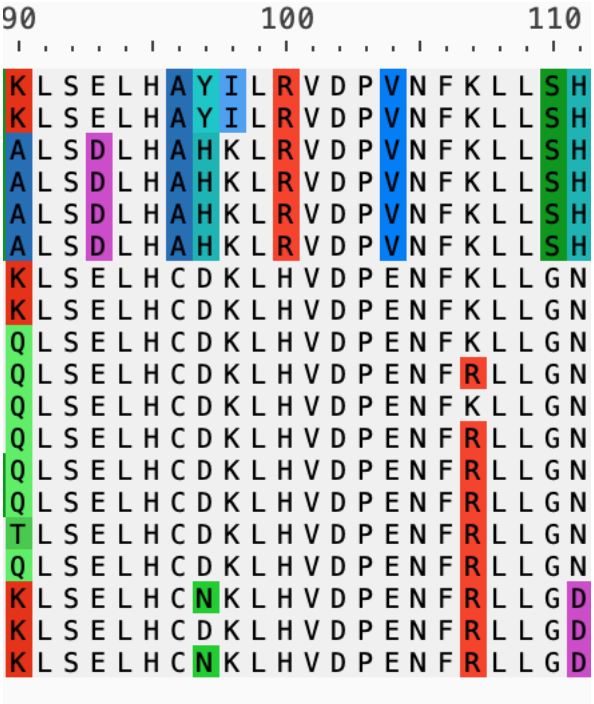


Figure 5: Figure 5 obtained while analysing the aligned protein sequences using Aliview [7]. Sequences were taken from NCBI Protein [1] and NCBI BLASTP [4], they were aligned using MAFFT [5] software.

3. Restructuring of the phylogeny trees

The sequences were put on IQTREE [8] for construction on phylogeny trees using 10000 as bootstrap values [9]. The imaging was obtained by FigTree [10].

Table 4: Model selection and the points by IQTREE [8][9].

The sequence	Model Selected	BIC Value
DNA	HKY+F+I	6102.527 +
Protein	Dayhoff+G4	3296.103 +

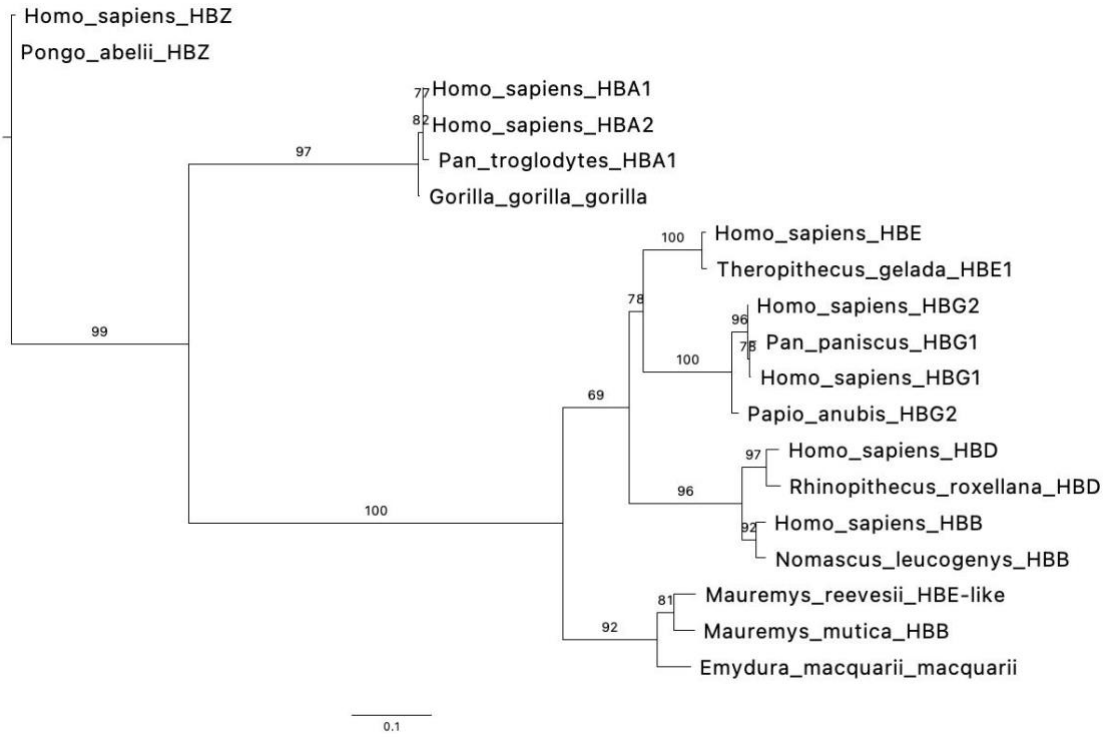


Figure 6: DNA phylogeny tree without re-rooting. The sequences were gathered from NCBI Nucleotide [2] and NCBI BLASTP [4], aligned with MAFFT [5] and tranalign [6]. The tree was reconstructed via IQTREE [8] software, using bootstrap value of 10000 [9], image obtained via Figtree [10]. The scores of bootstrap support values are shown on the branches (Emydura macquarii macquarii HBG1, Gorilla gorilla gorilla HBA).

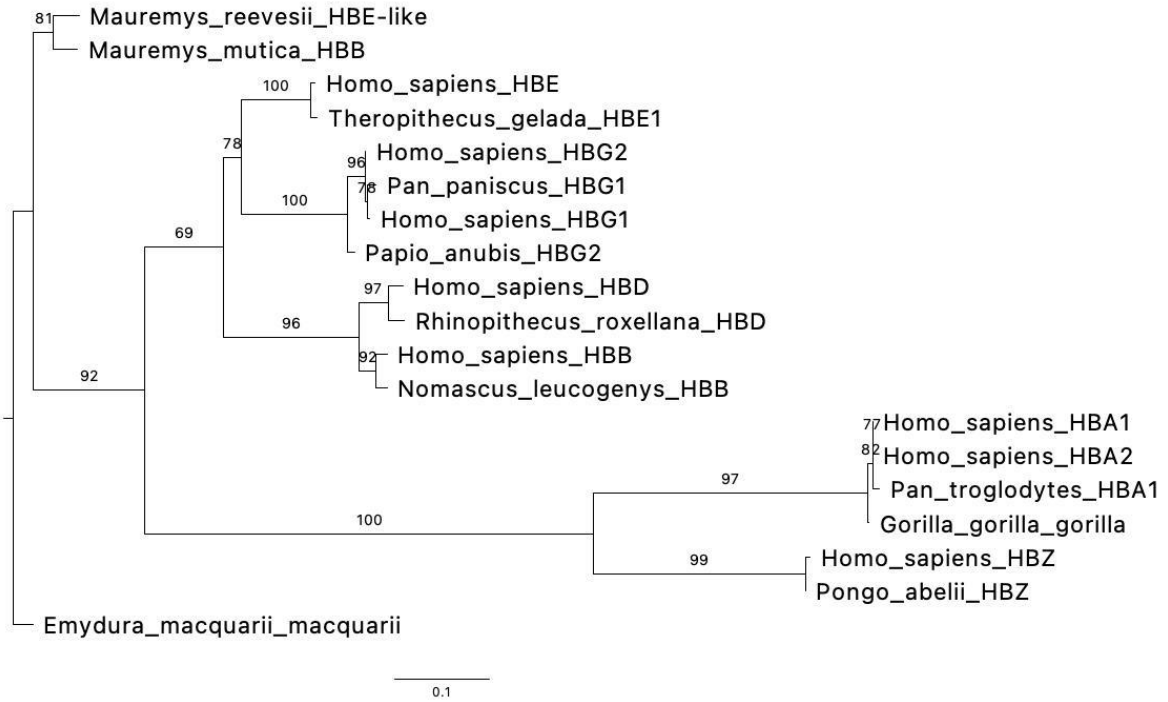


Figure 7: DNA phylogeny tree with re-rooting of outgroups. The sequences were gathered from NCBI Nucleotide [2] and NCBI BLASTP [4], aligned with MAFFT [5] and tranalign [6]. The tree was reconstructed via IQTREE [8] software, using bootstrap value of 10000 [9], image obtained via Figtree [10]. The scores of bootstrap support values are shown on the branches (Emydura macquarii macquarii HBG1, Gorilla gorilla gorilla HBA).

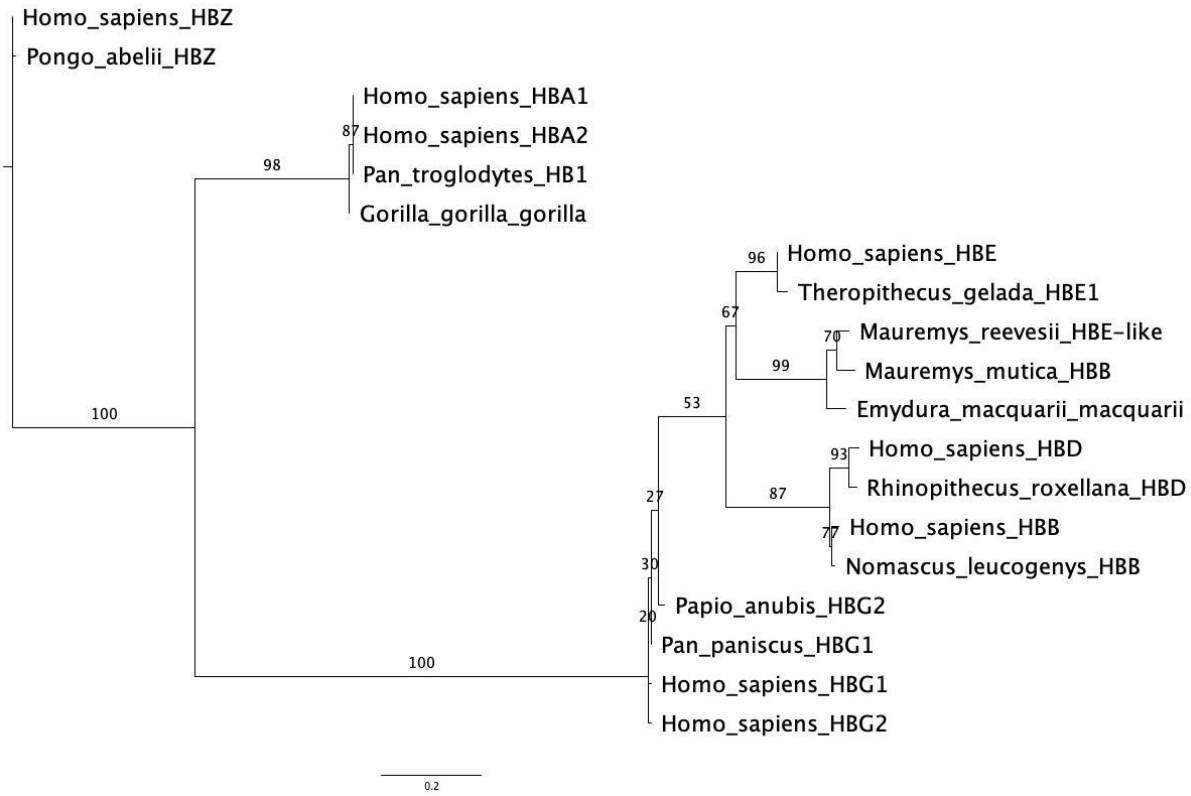


Figure 8: Protein phylogeny tree without re-rooting. The sequences were gathered from NCBI Protein [1] and NCBI BLASTP [4], aligned with MAFFT [5]. The tree was reconstructed via IQTREE [8] software, using bootstrap value of 10000 [9], image obtained via Figtree [10]. The scores of bootstrap support values are shown on the branches (Emydura macquarii macquarii HBG1, Pan troglodytes HBA1, Gorilla gorilla gorilla HBA).

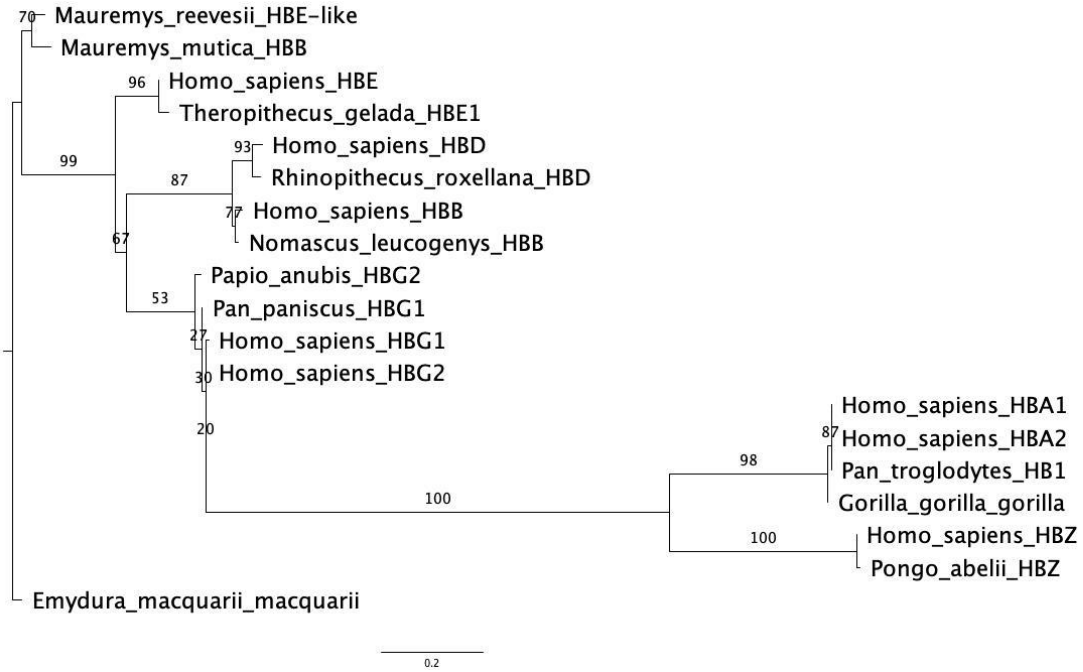


Figure 9: Protein phylogeny tree with re-rooting of outgroups. The sequences were gathered from NCBI Protein [1] and NCBI BLASTP [4], aligned with MAFFT [5]. The tree was reconstructed via IQTREE [8] software, using bootstrap value of 10000 [9], image obtained via Figtree [10]. The scores of bootstrap support values are shown on the branches (Emydura macquarii macquarii HBG1, Pan troglodytes HBA1, Gorilla gorilla gorilla HBA).

Discussion

Ingroups and Outgroups

In this paper, the ingroup was chosen from mammalian classes based on sequence similarity to each of the human hemoglobin proteins. The outgroups were selected from class Reptilia, and their sequence similarity to human globin proteins beta, delta, and gamma-1. The reason for this was that the reptiles and mammalians diverged from organisms that were in the Therapsida order which belonged to reptilians [11]. They were re-rooted on the trees as seen in Figure 7 and 9 accordingly via FigTree [10] software.

Phylogeny Trees and Aligned Sequences

Aligned Sequences

As stated in the paper written by Hardison in 2012 [12], the human hemoglobin genes are generally separated into two groups: *HBA* and *HBB* [12]. Moreover, the types of the globins are divided according to their relations to each gene due to duplication which means the gene has been [12]. *HBZ* gene and its protein zeta-globin are referred as alpha-like while epsilon, gamma and delta globins are referred as beta-like [12]. The difference in the sequences of these two groups could be seen after alignment on Aliview [7] in Figure 1, as alpha-like globins

have similar sequences at the sites 67-72 while beta-like globins have gaps at the said sites. Likewise, in Figure 2, it could be seen that at the sites 160-179, beta-globins have similar sequences while alpha-like globins have gaps. This could be a major factor in terms of determining their relations to each other. In Figure 3, the DNA sequences seem to align in majority of the sites while the grouping between them could still be seen.

When it comes to differentiation between the groups, Figure 4 shows that amino acids at the sites tend to align generally better with their own group (alpha-like and beta-like) [12] while aligning to a greater degree with same genes in different species such as *HBZ* in Homo sapiens and Pongo abelii. Moreover, it could also be seen that the outgroups have more amino acids in common with each other when they are compared to the mammalian sequences. Figure 5 is also supporting this statement.

The hemoglobin genes are found in clusters throughout the genome: MN, LCA, DS and LA [12]. The locations of the genes and their clusters changes across species [12]. Alpha-like hemoglobins (*HBA* and *HBZ*) are majorly located on the MN cluster and are stable [12]. On the other hand, beta-like hemoglobins are mostly conserved on DS locus [12]. Starting from 5' end the beta-like globins are located as follows: *HBE, HBG, HBD, HBB* [12]. It is predicted that the *HBB* and *HBD* genes share an ancestor gene created by duplication while another duplication event produced the ancestor of *HBG* and *HBE* genes [12]. This can be seen in Figure 7, as the branches suggest the grouping of *HBG* and *HBE* genes while suggesting the grouping of *HBB* and *HBD* genes.

Reconstructed Phylogeny Trees

For DNA sequences, the model selected by IQTREE [8] was HKY+F+I with BIC value of 6102.527 +, while for protein sequences the model was Dayhoff+G4 with BIC value of 3296.103 +. Both trees were re-rooted as this paper has an outgroup [13]. For both trees, without re-rooting, it could be seen that the outgroups were grouped together however, they were grouped closer to *HBE* and *HBD* genes and their products (Figure 6 and 8). *HBG* gene was grouped with *HBE* and *HBD* genes in the tree that was based on DNA sequences (Figure 6) however, its products were placed further away in the tree based on protein sequences (Figure 8), and it could be seen that *HBA* and *HBZ* genes and their proteins are not grouped together. These are no longer the case after re-rooting the trees based on outgroups (Figure 7 and 9). The support values for the branches overall seem to be higher in the DNA based trees compared to protein-based trees.

According to the paper written by Mao et al. (2022), In some mammals, loss of *HBB* gene caused *HBD* gene to function similar to *HBB* gene due to gene conversion [14]. This could explain the grouping of *HBD* and *HBB* genes together in the trees. Moreover, in same paper, it is mentioned that gene conversion had affected beta-like globins in mammals and reptiles in a higher rate and apart from each other [14]. This could explain the multiple branching of beta-like globins. In addition to that, it is seen on the trees that *HBG* genes show high support values on their branching while this is not the case for gamma-globins in protein trees. This means that the model was not confident in the branching [13], while in DNA trees, the value for this group is higher although it is still lower than the rest. This could be due to the high conversion rate of *HBG* genes in other mammalians that are not humans [14]. Moreover, it is seen that *HBG1* and *HBG2* genes from Homo sapiens was clustered together with *HBG1* gene from Pan paniscus while *HBG2* gene from Papio anubis was clustered slightly away from them (Figure 7). This could be due to the regular occurring of gene conversation in *HBG1* gene in mammals excluding humans [14].

Overall, I would consider the tree that was based on DNA sequences more reliable as it provides more detailed branching and is confirmed more by the literature. We could see the duplication event regarding beta-like genes in Figure 7 as mentioned beforehand which is not the case for the tree that was generated from the protein sequences. In addition to that it shows more confidence on branching due to the support values.

Further studies could include studying the mutations between *HBG* genes and their products to better understand the uncertainty of the software regarding the support values. Moreover, in mammals, some globin genes such as *HBG* and *HBD* are not conserved across the species while *HBE* is [12]. They are also not consistently present throughout the species and their presence changes [12]. On the cluster, the unstable genes (*HBG, HBD* and *HBN)* are in the middle [12]. The relationship between their location and stability through evolution could be studied. Moreover, as mentioned before the branching of *HBG2* of Papio Anubis apart from the rest of *HBG* genes could be studied.

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